

Enhancement One Narrative

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For the software design and engineering enhancement, I selected the Corner Grocer application. I developed this C++ program for my final project in the SNHU Programming Languages course, which I took two years ago. In the original version of this application, the program would begin by reading a text file containing purchase history data for the produce section of Corner Grocer. The program would load the name of each unique item into a vector and load the name of each unique item and how many times it had been purchased into a map. By interacting with a menu, the user could accomplish two tasks: view how many times a particular produce item was purchased or view a complete list of how many times each produce item was purchased. When viewing the complete list, the user could choose whether the number of times each item was purchased would be represented by digits or a histogram.

I chose to include this artifact in my portfolio because I can demonstrate growth in my skills and abilities in software development. Looking back on this program that I developed early in the SNHU Computer Science program, I found many opportunities for improvement. Therefore, in this enhancement I am demonstrating my ability to create a more robust program that follows coding best practices by incorporating additional input validation, error handling, and eliminating many hardcoded values to make future updates as seamless as possible.

While I am making these improvements to the code itself, the primary enhancement to this program is translating it from C++ into Python. Through this translation, I am showcasing my coding skills in multiple programming languages. Additionally, when put together with my enhancements in the data structures and algorithms and databases categories, which will involve storing produce purchase history information in a database instead of a text file, I will

demonstrate my ability to create a cohesive and fully functional software application designed for long-term use.

I did meet the course outcomes that I planned to meet for the software design and engineering enhancement. I implemented additional input validation and error handling methods to enhance the security of the program and protect against potential vulnerabilities such as injection attacks. I also initialized each variable with a value to eliminate the possibility of unexpected program behavior due to referencing a variable without a value assigned to it. These additions helped me meet course outcome five: developing a security mindset that anticipates malicious behavior and takes action to eliminate potential vulnerabilities and design flaws by writing secure code. I also met course outcome four by implementing additional components that add value to this application, such as additional input validation, error handling, and reducing hardcoded values. Finally, my in-line comments and this narrative demonstrate my ability to develop and deliver professional written communication, which shows progress in meeting course outcome two. I will continue meeting these course outcomes in one or both of my next enhancements.

During the process of translating the C++ program into Python, I came across a few challenges. The first challenge was that the translation process took longer than I expected. Because I have primarily been working with C++ and Java recently, I found myself consistently researching proper syntax as I was writing the code. This slowed the translation process quite a bit. Another challenge I faced was determining how to write a couple of the functions that interact with purchase history data because the database does not yet exist. For this enhancement, I wrote two additional functions that were not in my original pseudocode to return sample data.

This allowed me to fully write and test the rest of the code. In the final program, these functions will be replaced by database queries.

While working on this enhancement, I solidified my understanding of Python syntax and the unique features associated with this language, such as the method required to write a main function. I also learned how to set up a virtual environment and create a requirements text file in Visual Studio Code. The Visual Studio Code documentation for getting started with Python was an extremely helpful resource that I used to learn this process. The requirements text file will ensure that viewers of my ePortfolio will be able to download and run my code using the same dependencies I used to write the code. Additionally, this ensures that as a developer, I can work on this project in different environments if needed.